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Method for fabricating array substrate

Abstract

A method for fabricating an array substrate, the array substrate, a display panel, and a thin film transistor are provided. The thin film transistor includes a gate electrode, an active layer, a source electrode, a drain electrode, and an interlayer insulating layer. The active layer is disposed corresponding to the gate electrode. The source electrode and the drain electrode are disposed at both sides of the active layer and electrically connected to the active layer. The interlayer insulating layer is disposed between the active layer and the source electrode, and between the active layer and the drain electrode. The interlayer insulating layer is provided with step-shaped contact holes. The source electrode and the drain electrode are filled in the contact holes and electrically connected to the active layer.

Inventors: Gong; Jixiang (Hubei, CN), Zhang; Yixian (Hubei, CN), Xianyu; Wenxu (Hubei, CN)

Applicant: Wuhan China Star Optoelectronics Semiconductor Display Technology Co., Ltd. (Hubei, CN)

Family ID: 72566441

Assignee: Wuhan China Star Optoelectronics Semiconductor Display Technology Co., Ltd. (Wuhan, CN)

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Background/Summary

RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 16/978,786 filed on Sep. 8, 2020, which application is a National Phase of PCT Patent Application No. PCT/CN2020/097537 having International filing date of Jun. 22, 2020, which claims the benefit of priority of Chinese Patent Application No. 202010529346.6 filed on Jun. 11, 2020. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

(1) The present disclosure relates to the technical field of display, and particularly to a thin film transistor, an array substrate, a display panel, and a method for fabricating the array substrate.

(2) In current organic light-emitting diode (OLED) display devices, in order to achieve low power consumption, a mainstream technology is to use low-temperature poly-silicon (LTPS) in driving thin film transistors (TFTs) and switching thin film transistors as active layers. However, LTPS has a large carrier mobility and has a problem of a high leakage current. On this basis, low temperature polycrystalline-Si oxide (LTPO) technology was developed. LTPO comprises advantages of both LTPS and metal oxides, such as indium gallium zinc oxide, and has become a solution with fast response and low power consumption. However, because LTPO uses two types of active layers at same time, a number of photomasks required in a fabricating process increases, the fabricating process is more complicated, and cost also increases.

SUMMARY OF THE INVENTION

(3) In view of this, a purpose of the present disclosure is to provide a method for fabricating an array substrate that is simple to manufacture and low in cost, and can reduce a number of photomasks, and to provide the array substrate and a thin film transistor.

(4) The present disclosure provides a thin film transistor, comprising: a gate electrode; an active layer disposed corresponding to the gate electrode; a source electrode; a drain electrode, wherein the source electrode and the drain electrode are disposed at both sides of the active layer and electrically connected to the active layer; and an interlayer insulating layer disposed between the active layer and the source electrode, and between the active layer and the drain electrode, wherein the interlayer insulating layer is provided with step-shaped contact holes, and the source electrode and the drain electrode are filled in the contact holes and electrically connected to the active layer.

(5) In an embodiment, each of the contact holes is composed of a first contact hole and a second contact hole that communicate with each other, the first contact hole is disposed on a side of the second contact hole away from the active layer, a diameter of the first contact hole is greater than that of the second contact hole, the first contact hole has a first smooth rounded surface at its end away from the active layer, and the second contact hole has a second smooth rounded surface at its

end away from the active layer.

(6) In an embodiment, the second contact holes extend to the active layer.

(7) In an embodiment, the thin film transistor further comprises a light-shielding layer disposed below the active layer, and a buffer layer disposed between the light-shielding layer and the active layer. The second contact holes extend through the active layer into the buffer layer.

(8) The present disclosure further provides an array substrate comprising a substrate, a first thin film transistor, and a second thin film transistor. The first thin film transistor is disposed on the substrate and comprises: a first gate electrode; a first active layer disposed corresponding to the first gate electrode; a first source electrode; a first drain electrode, wherein the first source electrode and the first drain electrode are disposed at both sides of the first active layer and electrically connected to the first active layer; and a first interlayer insulating layer disposed between the first active layer and the first source electrode, and between the first active layer and the first drain electrode, wherein the first interlayer insulating layer is provided with step-shaped contact holes, and the first source electrode and the first drain electrode are filled in the contact holes and electrically connected to the first active layer. The second thin film transistor is disposed on the substrate and comprises: a second gate electrode; a second active layer disposed corresponding to the second gate electrode and on a side of the second active layer away from the substrate; a second source electrode; a second drain electrode, wherein the second source electrode and the second drain electrode are disposed at both sides of the second active layer and electrically connected to the second active layer; and a second interlayer insulating layer disposed between the second active layer and the first source electrode, and between the second active layer and the first drain electrode. The first drain electrode and the second source electrode are electrically connected through a first connection metal layer.

(9) In an embodiment, the array substrate further comprises a light-shielding layer disposed between the substrate and the first thin film transistor. The first source electrode and the light-shielding layer are electrically connected through a second connection metal layer.

(10) In an embodiment, the first active layer is made of low-temperature polysilicon, and the second active layer is made of a metal oxide semiconductor material.

(11) In an embodiment, each of the contact holes is composed of a first contact hole and a second contact hole that communicate with each other. The first contact hole is disposed on a side of the second contact hole away from the first active layer. A diameter of the first contact hole is greater than that of the second contact hole. The first contact hole has a first smooth rounded surface at its end away from the first active layer. The second contact hole has a second smooth rounded surface at its end away from the active layer.

(12) In an embodiment, the second contact holes extend to the first active layer.

(13) In an embodiment, the thin film transistor further comprises a light-shielding layer disposed below the first active layer, and a buffer layer disposed between the light-shielding layer and the first active layer. The second contact holes extend through the active layer into the buffer layer.

(14) The present disclosure further provides a method for fabricating an array substrate, comprising: providing a first substrate comprising a substrate, a first active layer, a first interlayer insulating layer, a second active layer, and a second interlayer insulating layer, wherein the first active layer and the second active layer are disposed on the substrate, the second active layer is disposed on a side of the first active layer away from the substrate, the first interlayer insulating layer is disposed between the first active layer and the second active layer, and the second interlayer insulating layer is disposed on the second active layer; coating a photoresist layer on the first substrate; patterning the photoresist layer using a halftone mask, wherein first through holes are formed at positions of the photoresist layer corresponding to both sides of the first active layer, and first blind holes are formed at positions of the photoresist layer corresponding to both sides of the second active layer; etching away the second interlayer insulating layer at positions of the first through holes to form first contact holes exposing the first interlayer insulating layer; ashing the

patterned photoresist layer, so that the first blind holes form second through holes; etching away the first interlayer insulating layer at positions of the first contact holes to form second contact holes communicating with the first contact holes and exposing the first active layer; etching away the second interlayer insulating layer at positions of the second through holes to form third contact holes exposing the second active layer; removing the photoresist layer; and forming a source/drain metal layer on the second interlayer insulating layer, wherein the source/drain metal layer comprises a first source electrode, a first drain electrode, a second source electrode, a second drain electrode, and a first connection metal layer, the first source electrode and the first drain electrode respectively fill the first contact holes and respectively fill the second contact holes communicating with the first contact holes, the second source electrode and the second drain electrode respectively fill the third contact holes, and the first drain electrode and the second source electrode are electrically connected through the first connection metal layer.

(15) In an embodiment, the second contact holes extend from the first interlayer insulating layer to the active layer.

(16) In an embodiment, the first substrate further comprises a first gate metal layer and a first light-shielding layer that are disposed on the substrate. The first gate metal layer is disposed corresponding to the first active layer. The first light-shielding layer is disposed below the first gate metal layer and the first active layer. The first gate metal layer comprises a light-shielding layer connection electrode electrically connected to the first light-shielding layer. In the steps of coating the photoresist layer on the first substrate and patterning the photoresist layer using the halftone mask, a third through hole is formed at a position of the photoresist layer corresponding to the light-shielding layer connection electrode. In the step of etching away the second interlayer insulating layer at positions of the first through holes to form the first contact holes, the second interlayer insulating layer at a position of the third through hole is etched away to form a fourth contact hole exposing the first interlayer insulating layer. In the steps of etching away the first interlayer insulating layer at positions of the first contact holes to form the second contact holes communicating with the first contact holes and etching away the second interlayer insulating layer at positions of the second through holes to form the third contact holes, the first interlayer insulating layer at a position of the fourth contact hole is etched away to form a fifth contact hole exposing the light-shielding layer connection electrode. In the steps of removing the photoresist layer and forming the source/drain metal layer on the second interlayer insulating layer, the source/drain metal layer further comprises a second connection metal layer, wherein the second connection metal layer fills the fourth contact hole and the fifth contact hole communicating with the fourth contact hole, and is electrically connected to the first source electrode.

(17) In an embodiment, the photoresist layer is made of a positive photoresist, and the halftone mask comprises a plurality of light-transmitting regions having a light transmittance of 100%, a plurality of semi-light-transmitting regions having a light transmittance of 40%-70%, and a plurality of non-light-transmitting regions having a light transmittance of 0%.

(18) In an embodiment, the first active layer is made of low-temperature polysilicon, and the second active layer is made of a metal oxide semiconductor material.

(19) Compared with the prior art, by using a half-tone mask and only one photomask process, a method for fabricating an array substrate of the present disclosure forms a plurality of contact holes for electrically connecting a first thin-film transistor and a second thin-film transistor. This reduces photomask processes, improves production efficiency, and reduces costs. In addition, in an array substrate fabricated by the method for fabricating the array substrate of the present disclosure, ports of contact holes of a first source electrode and a first drain electrode have smooth arcs, which can effectively prevent wire breakage caused by holes being too deep.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In order to more clearly illustrate the technical solutions in the embodiments of the present disclosure, a brief description of accompanying drawings used in the description of the embodiments of the present disclosure will be given below. Obviously, the accompanying drawings in the following description are merely some embodiments of the present disclosure. For those skilled in the art, other drawings may be obtained from these accompanying drawings without creative labor.

(2) FIG. 1(a), FIG. 1(b), FIG. 1(c), FIG. 1(d), FIG. 1(e), FIG. 1(f) and FIG. 1(g) are schematic flowcharts of a method for fabricating an array substrate according to a first embodiment of the present disclosure.

(3) FIG. 2 is a schematic cross-sectional view of a display panel according to a second embodiment of the present disclosure.

(4) FIG. 3 is a schematic diagram of contact holes of the display panel of FIG. 2.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(5) Technical solutions of the present disclosure will be clearly and completely described below in conjunction with accompanying drawings in embodiments of the present disclosure. It is apparent that the described embodiments are merely a part of the embodiments of the present disclosure and not all embodiments. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative labor are within claimed scope of the present disclosure.

(6) Please refer to FIG. 1(a) to FIG. 1(g), a first embodiment of the present disclosure provides a method for fabricating an array substrate, comprising the following steps.

(7) S1: please refer to FIG. 1(a), providing a first substrate **100a**. The first substrate **100a** comprises a substrate **10**, a first active layer **21**, a first gate metal layer **22**, a first interlayer insulating layer **30**, a second active layer **41**, a second gate metal layer **42**, and a second interlayer insulating layer **50**. The first active layer **21** and the first gate metal layer **22** are disposed on the substrate **10**. The first interlayer insulating layer **30** is disposed on the first active layer **21** and the first gate metal layer **22**. The second active layer **41** and the second gate metal layer **42** are disposed on the first interlayer insulating layer **30**. The second interlayer insulating layer **50** is disposed on the second active layer **41** and the second gate metal layer **42**.

(8) The substrate **10** is a transparent substrate, and may be made of a flexible material such as polyimide.

(9) The first active layer **21** is disposed on a side of the first gate metal layer **22** close to the substrate **10**. A first gate insulating layer is disposed between the first active layer **21** and the first gate metal layer **22**. The second active layer **41** is disposed on a side of the second gate metal layer **42** close to the substrate **10**. A second gate insulating layer is disposed between the second active layer **41** and the second gate metal layer **42**. Therefore, thin film transistors formed subsequently are bottom-gate thin film transistors. However, in the present disclosure, types of thin film transistors formed subsequently are not limited, and the thin film transistors formed subsequently may be bottom-gate thin film transistors or top-gate thin film transistors. In a top-gate thin film transistor, the first gate metal layer **22** is disposed on a side of the first active layer **21** close to the substrate **10**, and the second gate metal layer **42** is disposed on a side of the second active layer **41** close to the substrate **10**.

(10) In this embodiment, the first active layer **21** is made of low-temperature polysilicon such as N-type doped low-temperature polysilicon. The second active layer **41** is made of a metal oxide semiconductor material, such as indium gallium zinc oxide (IGZO), indium gallium zinc tin oxide (IGZTO), indium zinc oxide (IZO), gallium indium oxide (IGO), indium gallium zinc oxide (IGZO), indium gallium tin oxide (IGTO), indium zinc tin oxide (IZTO), and indium tin oxide (ITO).

- (11) The first gate metal layer **22** comprises a first gate electrode **221**, a second light-shielding layer **222**, and a light-shielding layer connection electrode **223**. The first gate electrode **221** is disposed corresponding to the first active layer **21**. The second light-shielding layer **222** and the light-shielding layer connection electrode **223** are disposed on both sides of the first gate electrode **221**, respectively.
- (12) The second gate metal layer **42** comprises a second gate electrode **422**. The second gate electrode **422** is disposed corresponding to the second active layer **41**. The second light-shielding layer **222** is configured to shield the second gate electrode **422** from light.
- (13) In addition, the first substrate **100a** further comprises a first light-shielding layer **60** disposed on the substrate **10**. The first light-shielding layer **60** is disposed below the first gate metal layer **22** and the first active layer **21**. The light-shielding layer connection electrode **223** is electrically connected to the first light-shielding layer **60** through a through hole penetrating a first gate electrode light-shielding layer and a buffer layer. The first light-shielding layer **60** is configured to shield the first gate electrode **221** from light. It can be understood that a buffer layer and the like may also be disposed between the substrate **10** and the first light-shielding layer **60**.
- (14) The first gate metal layer **22**, the second gate metal layer **42**, and the first light-shielding layer **60** may be made of tantalum (Ta), tungsten (W), molybdenum (Mo), aluminum (Al), titanium (Ti), copper-niobium (Cu—Nb) alloy, or the like, and may also be made of a stack of copper (Cu) and molybdenum (Mo), a stack of copper (Cu) and molybdenum-titanium (Mo—Ti) alloys, a stack of copper (Cu) and titanium (Ti), a stack of aluminum (Al) and molybdenum (Mo), a stack of molybdenum (Mo) and tantalum (Ta), a stack of molybdenum (Mo) and tungsten (W), a stack of molybdenum (Mo)-aluminum (Al)-molybdenum (Mo), or the like.
- (15) The first interlayer insulating layer **30**, the second interlayer insulating layer **50**, the first gate insulating layer, and the second gate insulating layer may be made of SiO_x, SiN_x, a stack of SiN_x and SiO_x, or the like.
- (16) S2: please refer to FIG. **1(b)** and FIG. **1(c)**, coating a photoresist layer **200** on the first substrate **100a**, and patterning the photoresist layer **200** using a halftone mask **300**. The step of patterning comprises exposing and developing the photoresist layer **200** using the halftone mask **300**. By exposing and developing, two first through holes **201** are formed at positions of the photoresist layer **200** corresponding to both sides of the first active layer **21**, and two first blind holes **202** are formed at positions of the photoresist layer **200** corresponding to both sides of the second active layer **41**. In other words, the two first through holes **201** are formed at positions of the photoresist layer **200** corresponding to predetermined source/drain electrodes on the first active layer **21**, and the two first blind holes **202** are formed at positions of the photoresist layer **200** corresponding to predetermined source/drain electrodes on the second active layer **41**. A depth of the first blind holes **202** occupies 40%-70% of a thickness of the photoresist layer **300**.
- (17) In this step, a third through hole **203** is also formed at a position of the photoresist layer **200** corresponding to the light-shielding layer connection electrode **223**.
- (18) The photoresist layer **200** may be made of a positive photoresist or a negative photoresist. In this embodiment, the photoresist layer **200** is a positive photoresist layer. The halftone mask **300** comprises a plurality of light-transmitting regions **301**, a plurality of semi-light-transmitting regions **302**, and a plurality of non-light-transmitting regions **303**. The light-transmitting regions **301** have a light transmittance of 100%. The semi-light-transmitting regions **302** have a light transmittance of 40%-70%. The non-light-transmitting regions **303** have a light transmittance of 0%.
- (19) S3: please refer to FIG. **1(c)** and FIG. **1(d)**, etching away the second interlayer insulating layer **50** at positions of the first through holes **201** to form first contact holes **51**, and at a position of the third through hole **203** to form a fourth contact hole **53**. The first contact holes **51** and the fourth contact hole **53** expose the first interlayer insulating layer **30**.
- (20) S4: please refer to FIG. **1(d)** and FIG. **1(e)**, ashing the patterned photoresist layer **200**, so that

the first blind holes **202** form second through holes **202a**. The step of ashing uses oxygen or a gas comprising oxygen. The step of ashing thins the photoresist layer **200**, so that the first blind holes **202** form the second through holes **202a**.

(21) **S5**: please refer to FIG. **1(f)**, etching away the first interlayer insulating layer **30** at positions of the first contact holes **51** to form second contact holes **31**, etching away the second interlayer insulating layer **50** at positions of the second through holes **202a** to form third contact holes **52**, and etching away the first interlayer insulating layer **30** at a position of the fourth contact hole **53** to form a fifth contact hole **32**. The second contact holes **31** expose the first active layer **21**. The third contact holes **52** expose the second active layer **41**. The fifth contact hole **32** exposes the light-shielding layer connection electrode **223**.

(22) In an embodiment, the second contact holes **31** expose an upper surface of the first active layer **21**. In another embodiment, the second contact holes **31** extend from the first interlayer insulating layer **30** to the first active layer **21**. The description “extend . . . to the first active layer **21**” may be “penetrate a part of the first active layer **21**” or “penetrate the first active layer **21**”. The second contact holes **31** may also penetrate the first active layer **21** and a buffer layer below it, as long as the second contact holes **31** are not short-circuited with the first light-shielding layer **60**. In other embodiments of the present disclosure, the second contact holes **31** penetrate the interlayer insulating layer **30** and the first gate insulating layer to expose the first active layer **21**.

(23) **S6**: please refer to FIG. **1(g)**, removing the photoresist layer **200**, and forming a source/drain metal layer **70** on the second interlayer insulating layer **50** to form a first thin film transistor **T1** and a second thin film transistor **T2**, thereby obtaining the array substrate **100**.

(24) The source/drain metal layer **70** comprises a first source electrode **71**, a first drain electrode **72**, a second source electrode **73**, a second drain electrode **74**, a first connection metal layer **75**, and a second connection metal layer **76**. The first source electrode **71** and the first drain electrode **72** respectively fill the first contact holes **51**, and respectively fill the second contact holes **31** communicating with the first contact holes **51**. The source/drain metal layer **70** and the first active layer **21** are electrically connected in the second contact holes **31**. That is, side contacts. The second source electrode **73** and the second drain electrode **74** respectively fill the third contact holes **52**. The first drain electrode **72** and the second source electrode **73** are electrically connected through the first connection metal layer **75**. The second connection metal layer **76** fills the fourth contact hole **53** and the fifth contact hole **32** communicating with the fourth contact hole **53**, and is electrically connected to the first source electrode **71**. The first light-shielding layer **60** is electrically connected to the first source electrode **71** through the light-shielding layer connection electrode **223** and the second connection metal layer **76**. Therefore, an electric field caused by ions of layers below low-temperature polysilicon can be shielded, which makes electrical properties of thin film transistors are more stable.

(25) In addition, in the array substrate **100** fabricated by the aforementioned method, the second interlayer insulating layer **50** at an end of the first contact hole **51** away from the first interlayer insulating layer **30** has a first smooth rounded surface **51a**, and the first interlayer insulating layer **30** at an end of the second contact hole **31** close to the second interlayer insulating layer **50** has a second smooth rounded surface **31a**. The first smooth rounded surface **51a** and the second smooth rounded surface **31a** have smooth arcs instead of obtuse angle shapes formed by the two etchings. When the source/drain metal layer **70** covers the first contact hole **51** and the second contact hole **31**, these smooth arcs can effectively prevent wire breakage caused by holes being too deep.

(26) In other embodiments of the present disclosure, the array substrate **100** further comprises a third active layer disposed on the second interlayer insulating layer, and a fourth interlayer insulating layer disposed on the third active layer. In order to electrically connect the second active layer **41** and the third active layer, the array substrate **100** can be fabricated in the same manner. Specifically, a second semi-light-transmitting region is added to the halftone mask **300**. A light transmittance of the semi-light-transmitting region is less than that of the semi-light-transmitting

region **302**. Through the above steps of etching, ashing, and etching, contact holes between the second active layer **41** and the third active layer can be obtained. The second active layer **41** and the third active layer can be electrically connected through the source/drain metal layer **70**. In other embodiments of the present disclosure, materials of the first active layer **21** and the second active layer **41** are not limited. As long as the array substrate **100** has the first active layer **21** and the second active layer **41** in different layers, the method for fabricating the array substrate of the present disclosure can be used to connect the first thin film transistor T1 and the second thin film transistor T2.

(27) Please refer to FIG. 2 and FIG. 3, a second embodiment of the present disclosure further provides a display panel **1** comprising an array substrate **100** and a plurality of light emitting portions **400** disposed on the array substrate **100**.

(28) The array substrate **100** can be fabricated by the method of the first embodiment.

(29) The array substrate **100** comprises a substrate **10**, a first thin film transistor T1, and a second thin film transistor T2. The first thin film transistor T1 and the second thin film transistor T2 are disposed on the substrate **10**. The first thin film transistor T1 is disposed on a side of the second thin film transistor T2 close to the substrate **10**.

(30) The substrate **10** is a transparent substrate, and may be made of a flexible material such as polyimide.

(31) The first thin film transistor T1 comprises a first gate electrode **221**, a first active layer **21**, a first source electrode **71**, a first drain electrode **72**, and a first interlayer insulating layer **30**. The first active layer **21** is disposed corresponding to the first gate electrode **221**. A first gate insulating layer is further disposed between the first gate electrode **221** and the first active layer **21**. The first source electrode **71** and the first drain electrode **72** are disposed at both sides of the first active layer **21** and are electrically connected to the first active layer **21**. The first interlayer insulating layer **30** is disposed between the first active layer **21** and the first source **71**, and between the first active layer **21** and the first drain **72**. Step-shaped contact holes **30a** are disposed in the first interlayer insulating layer **30**. The first source electrode **71** and the first drain electrode **72** are respectively filled in the contact holes **30a** and electrically connected to the first active layer **21**. Each of the contact holes **30a** is composed of a first contact hole **51** and a second contact hole **31** that communicate with each other. The first contact hole **51** is disposed on a side of the second contact hole **31** away from the first active layer **21**. A diameter of the first contact hole **51** is greater than that of the second contact hole **31**.

(32) Please refer to FIG. 1(a) to FIG. 1(g), the first contact holes **51** are formed in a second interlayer insulating layer **50** at positions corresponding to both sides of the first active layer **21**. The second contact holes **31** are formed in the first interlayer insulating layer **30**. The second contact holes **31** expose the first active layer **21**.

(33) In an embodiment, the second contact holes **31** extend to the first active layer **21**. The first source electrode **71** and the first drain electrode **72** respectively fill the first contact holes **51**, and respectively fill the second contact holes **31** communicating with the first contact holes **51**. The source/drain metal layer **70** and the first active layer **21** are electrically connected in the second contact holes **31**. That is, side contacts.

(34) The first contact hole **51** has a first smooth rounded surface **51a** at its end away from the first active layer **21**. The second contact hole **31** has a second smooth rounded surface **31a** at its end away from the first active layer **21**. The first smooth rounded surface **51a** and the second smooth rounded surface **31a** have smooth arcs instead of obtuse angle shapes formed by the two etchings. When the source/drain metal layer **70** covers the first contact hole **51** and the second contact hole **31**, these smooth arcs can effectively prevent wire breakage caused by holes being too deep.

(35) In an embodiment, the array substrate **100** further comprises a first light-shielding layer **60** disposed between the substrate **10** and the first thin film transistor T1, and a buffer layer **80** disposed between the first light-shielding layer **60** and the first active layer **21**. The first light-

shielding layer **60** is configured to shield the first gate electrode **221** from light. The second contact holes **31** extend through the first active layer **21** into the buffer layer **80**. The first source electrode **71** and the first light-shielding layer **60** are electrically connected through a second connection metal layer **76**. The array substrate **100** further comprises a light-shielding layer connection electrode **223** disposed on a same layer as the first gate electrode **221**. The first light-shielding layer **60** is electrically connected to the first source electrode **71** through the light-shielding layer connection electrode **223** and the second connection metal layer **76**. Therefore, an electric field caused by ions of layers below low-temperature polysilicon can be shielded, which makes electrical properties of thin film transistors are more stable.

(36) The second thin film transistor T2 is disposed on the first interlayer insulating layer **30**. The second thin film transistor T2 comprises a second gate electrode **422**, a second active layer **41**, a second source electrode **73**, a second drain electrode **74**, and the second interlayer insulating layer **50**. The second gate electrode **422** is disposed corresponding to the second active layer **421** and on a side of the second active layer **41** away from the substrate **10**. A second gate insulating layer is further disposed between the second gate electrode **422** and the second active layer **41**. The second source electrode **73** and the second drain electrode **74** are disposed at both sides of the second active layer **41** and electrically connected to the second active layer **41**. The second interlayer insulating layer **50** is disposed between the second active layer **41** and the first source electrode **73**, and between the second active layer **41** and the first drain electrode **72**. The first drain electrode **72** and the second source electrode **73** are electrically connected through a first connection metal layer **75**.

(37) The array substrate **100** may further comprise a second light-shielding layer **222** between the second thin film transistor T2 and the substrate **10**. The second light-shielding layer **222** is configured to shield the second gate electrode **422** from light. The second light-shielding layer **222** and the light-shielding layer connection electrode **223** are disposed on both sides of the first gate electrode **221**, respectively.

(38) Please refer to FIG. 1(a) to FIG. 1(g), third contact holes **52** are formed at positions of the second interlayer insulating layer **50** corresponding to both sides of the second active layer **41**. The third contact holes **52** expose the second active layer **41**. A fourth contact hole **53** is formed at a position of the second interlayer insulating layer **50** corresponding to the light-shielding layer connection electrode **223**. The fourth contact hole **53** exposes the first interlayer insulating layer **30**. A fifth contact hole **32** is formed at a position of the first interlayer insulating layer **30** corresponding to the fourth contact hole **53**. The fifth contact hole **32** exposes the light-shielding layer connection electrode **223**. The fourth contact hole **53** communicates with the fifth contact hole **32**. The second source electrode **73** and the second drain electrode **74** respectively fill the third contact holes **52**. The second connection metal layer **76** fills the fourth contact hole **53** and the fifth contact hole **32** communicating with the fourth contact hole **53**, and is electrically connected to the first source electrode **71**.

(39) The first active layer **21** is disposed on a side of the first gate electrode **221** close to the substrate **10**. The second active layer **41** is disposed on a side of the second gate electrode **422** close to the substrate **10**. That is, the first thin film transistor T1 and the second thin film transistor T2 of this embodiment are both bottom-gate thin film transistors. However, in the present disclosure, types of thin film transistors formed subsequently are not limited, and the thin film transistors formed subsequently may be bottom-gate thin film transistors or top-gate thin film transistors. In this embodiment, the first active layer **21** is made of low-temperature polysilicon such as N-type doped low-temperature polysilicon. The second active layer **41** is made of a metal oxide semiconductor material, such as indium gallium zinc oxide (IGZO), indium gallium zinc tin oxide (IGZTO), indium zinc oxide (IZO), gallium indium oxide (IGO), indium gallium zinc oxide (IGZO), indium gallium tin oxide (IGTO), indium zinc tin oxide (IZTO), and indium tin oxide (ITO).

(40) The light emitting portions 400 may be arranged in an array. The light emitting portions 400 may be, for example, organic light emitting diode light emitting portions, micro light emitting diode chips, or the like. Each of the organic light emitting diode light emitting portions may comprise an anode layer disposed on a driving circuit layer, and a hole injection layer, a hole transport layer, an electron transport layer, an organic light emitting material layer, an electron transport layer, an electron injection layer, and a cathode layer sequentially stacked on the anode layer. The anode layer is electrically connected to the first drain electrode 71.

(41) Compared with the prior art, by using a half-tone mask and only one photomask process, a method for fabricating an array substrate of the present disclosure forms a plurality of contact holes for electrically connecting a first thin-film transistor and a second thin-film transistor. This reduces photomask processes, improves production efficiency, and reduces costs. In addition, in an array substrate fabricated by the method for fabricating the array substrate of the present disclosure, ports of contact holes of a first source electrode and a first drain electrode have smooth arcs, which can effectively prevent wire breakage caused by holes being too deep.

(42) The embodiments of the present disclosure are described in detail above. The present disclosure uses specific examples to describe principles and embodiments of the present invention. The above description of the embodiments is only for helping to understand the present invention. Furthermore, those skilled in the art may make modifications to the specific embodiments and applications according to ideas of the present invention. In conclusion, the present specification should not be construed as a limitation to the present invention.

Claims

1. A method for fabricating an array substrate for a display panel, comprising: providing a first substrate comprising a substrate, a first active layer, a first interlayer insulating layer, a second active layer, and a second interlayer insulating layer, wherein the first active layer and the second active layer are disposed on the substrate, the second active layer is disposed on a side of the first active layer away from the substrate, the first interlayer insulating layer is disposed between the first active layer and the second active layer, and the second interlayer insulating layer is disposed on the second active layer; coating a photoresist layer on the first substrate; patterning the photoresist layer using a halftone mask, wherein two first through holes are formed at positions of the photoresist layer corresponding to both sides of the first active layer, and two first blind holes are formed at positions of the photoresist layer corresponding to both sides of the second active layer; etching away the second interlayer insulating layer at positions of the first through holes to form first contact holes exposing the first interlayer insulating layer; ashing the patterned photoresist layer, so that the first blind holes form second through holes; etching away the first interlayer insulating layer at positions of the first contact holes to form second contact holes communicating with the first contact holes and extending in or through the first active layer; etching away the second interlayer insulating layer at positions of the second through holes to form third contact holes exposing the second active layer; removing the photoresist layer; and forming a source/drain metal layer on the second interlayer insulating layer, wherein the source/drain metal layer comprises a first source electrode, a first drain electrode, a second source electrode, a second drain electrode, and a first connection metal layer, the first source electrode and the first drain electrode respectively fill the first contact holes, respectively fill the second contact holes communicating with the first contact holes, and extend in or through and contact the first active layer such that the first source electrode and the first drain electrode are electrically connected to the first active layer, the second source electrode and the second drain electrode respectively fill the third contact holes, and the first drain electrode and the second source electrode are electrically connected through the first connection metal layer.

2. The method for fabricating the array substrate for a display panel according to claim 1, wherein

- the first contact hole has a first smooth rounded surface at its end away from the first active layer.
3. The method for fabricating the array substrate for a display panel according to claim 1, wherein the third contact holes are formed at positions of the second interlayer insulating layer corresponding to both sides of the second active layer and expose the second active layer, and the second source electrode and the second drain electrode are filled in the third contact holes.
4. The method for fabricating the array substrate for a display panel according to claim 1, wherein the second contact holes extend from the first interlayer insulating layer to the active layer.
5. The method for fabricating the array substrate for a display panel according to claim 1, wherein: the first substrate further comprises a first gate metal layer and a first light-shielding layer that are disposed on the substrate, the first gate metal layer is disposed corresponding to the first active layer, the first light-shielding layer is disposed below the first gate metal layer and the first active layer, the first gate metal layer comprises a light-shielding layer connection electrode electrically connected to the first light-shielding layer; in the steps of coating the photoresist layer on the first substrate and patterning the photoresist layer using the halftone mask, a third through hole is formed at a position of the photoresist layer corresponding to the light-shielding layer connection electrode; in the step of etching away the second interlayer insulating layer at positions of the first through holes to form the first contact holes, the second interlayer insulating layer at a position of the third through hole is etched away to form a fourth contact hole exposing the first interlayer insulating layer; in the steps of etching away the first interlayer insulating layer at positions of the first contact holes to form the second contact holes communicating with the first contact holes and etching away the second interlayer insulating layer at positions of the second through holes to form the third contact holes, the first interlayer insulating layer at a position of the fourth contact hole is etched away to form a fifth contact hole exposing the light-shielding layer connection electrode; and in the steps of removing the photoresist layer and forming the source/drain metal layer on the second interlayer insulating layer, the source/drain metal layer further comprises a second connection metal layer, wherein the second connection metal layer fills the fourth contact hole and the fifth contact hole communicating with the fourth contact hole, and is electrically connected to the first source electrode.
6. The method for fabricating the array substrate for a display panel according to claim 1, wherein the second contact hole has a second smooth rounded surface at its end away from the first active layer.
7. The method for fabricating the array substrate for a display panel according to claim 6, wherein the photoresist layer is made of a positive photoresist, and the halftone mask comprises: a plurality of light-transmitting regions having a light transmittance of 100%; a plurality of semi-light-transmitting regions having a light transmittance of 40%-70%; and a plurality of non-light-transmitting regions having a light transmittance of 0%.
8. The method for fabricating the array substrate for a display panel according to claim 1, wherein the first active layer is made of low-temperature polysilicon, and the second active layer is made of a metal oxide semiconductor material.
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